Investment Portfolio Risk & Scenario Analysis

Vertex Portfolio Risk Lab ● Srujana Tech Fest 2026 | Problem F-06

Project: Vertex Portfolio Risk Lab (F-06)

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1. Executive Summary

Modern wealth management faces a fundamental paradox: individual asset diversification does not guarantee risk immunity. Portfolios often conceal severe shared vulnerabilities across systemic market factors, interest-rate shocks, currency shifts, and cross-sector contagion that traditional static metrics fail to expose. The Vertex Portfolio Risk Lab was engineered to solve this challenge for Problem F-06 by providing an interactive, full-stack quantitative decision-support environment that transforms complex multi-asset risk dynamics into transparent, inspectable scenarios.

Our solution unifies a high-performance Python/FastAPI quantitative analytical engine with a modern, responsive React/TypeScript dashboard. Key capabilities include multi-factor scenario stress testing, graph-based shock contagion modeling across supply and credit dependencies, analytical reverse stress testing (solving for failure thresholds), Herfindahl-Hirschman concentration analytics, and automated investor goal-conflict detection. Every risk calculation operates deterministically and explainably, providing advisors and investors with clear mathematical provenance rather than opaque black-box predictions.

2. Problem Statement & Requirements Coverage

The solution comprehensively implements all three core directives stipulated in Hackathon Problem F-06:

- **1. Portfolio & Risk Factor Accounting:** Incorporates granular asset allocations, historical volatilities, asset class categories, liquidity tags, investment horizons, and empirical correlation matrices.
- **2. Multi-Dimensional Scenario Simulation:** Parametric real-time evaluation of equity market collapses (-50% to +20%), interest rate adjustments (-500bps to +500bps), sector-specific disruptions, and currency foreign-exchange shifts.
- **3. Decision Support & Guardrails:** Provides automated detection of conflicts between investor horizon/liquidity and risk profile, computes defensive benchmark rebalancing comparisons, and strictly demarcates historical baselines from hypothetical forward projections.

3. System Architecture & Technical Stack

Vertex Portfolio Risk Lab is architected as a modular, decoupled client-server platform optimized for fast calculation throughput, strict type safety, and responsive user interaction.

Layer	Technology Stack	Key Architectural Role
Frontend Dashboard	React 18, TypeScript, Vite, TailwindCSS	Interactive visualization, shock controls, correlation heatmaps, projection cones
API Backend Server	Python 3.14, FastAPI, Uvicorn, Pydantic v2	High-throughput asynchronous REST API, OpenAPI docs, strict schema validation
Quantitative Engine	NumPy, SciPy, Math (backend/engine/quant.py)	Parametric VaR, CVaR, Ulcer Index, HHI concentration, reverse stress solvers
Contagion Network	Directed Graph Engine (backend/engine/contagion.py)	Multi-stage shock transmission across sector and counterparty dependencies
Investor Guardrails	Rule-based AI Advisor (backend/engine/ai_agent.py)	NLP scenario parsing, horizon/liquidity conflict verification, defensive ticket generator

4. Core Features & Capabilities

A. Pre-Configured Portfolio Archetypes

To enable rigorous demonstration across contrasting investor scenarios, the system provides pre-configured, validated multi-asset portfolios:

- **Aggressive Growth (IT Concentration):** Heavy concentration in tech equities (TCS, INFY) with ancillary banking and real-estate debt, demonstrating hidden sector vulnerabilities.

- **Conservative Retiree (Interest-Rate Sensitive):** Dominated by government and corporate debt with real-estate holdings, highly exposed to monetary policy tightening and interest-rate hikes.
- **All-Weather Balanced:** Strategically diversified mix spanning equities, sovereign bonds, gold, banking, and consumer defensives.

B. Multi-Factor Stress Testing & Real-Time Impact Analysis

Users can dynamically adjust market conditions via real-time sliders or natural-language prompts. The engine instantaneously computes:

- Total capital gain/loss, stressed portfolio valuation, and relative drawdown percentage.
- Asset allocation and Herfindahl-Hirschman concentration scores with sector risk multipliers.
- Dynamic correlation heatmap illustrating cross-asset movement and clustering risks.
- 10th-to-90th percentile stochastic dispersion cone alongside historical performance baselines.

C. Contagion Network Simulation

Real-world financial crises do not occur in isolation; shocks cascade through supply chains, credit markets, and liquidity channels. The Vertex contagion engine models these dependencies as a directed propagation graph. When an initial shock hits a specific sector (e.g., Banking or IT), the engine simulates 1st, 2nd, and 3rd order transmission effects with calibrated dampening factors and transmission lags.

D. Reverse Stress Testing & Fragility Discovery

Instead of asking 'What happens if the market drops 10%?', the reverse stress testing module answers: 'What magnitude of market or sector shock would breach a 25% portfolio loss threshold?' The module applies numerical optimization to solve for critical vulnerability inflection points and identifies the most fragile assets contributing to catastrophic drawdowns.

E. Automated Decision Guardrails & Defensive Allocation

The system acts as an intelligent financial guardrail by evaluating investor inputs against established suitability heuristics:

- **Horizon Conflict Detection:** Flags portfolios whose equity volatility contradicts a short (< 3 year) investment horizon.
- **Liquidity Safety Check:** Warns when illiquid holdings exceed designated emergency redemption thresholds.
- **Defensive Benchmark Alternative:** Generates a balanced, defensive alternative allocation and calculates the explicit opportunity cost and drawdown protection delta.

5. Quantitative & Mathematical Framework

The Vertex engine is built on robust financial economics principles. All calculations are inspectable and reproducible:

- **Asset-Level Shock Propagation:** Evaluated as: $\Delta R_i = \beta_i \cdot \Delta M + S_{sector} + D_i \cdot \Delta r + C_{fx} \cdot \Delta FX + T_{contagion}$, where β is asset beta, D is duration, and T represents modeled network transmission.
- **Concentration Index (HHI):** Measured as $HHI = \Sigma (w_i)^2$. Concentration penalties are automatically triggered when any single asset or sector exceeds 30% aggregate weight.
- **Downside Tail Risk Metrics:** Employs parametric Value-at-Risk (VaR_{95}) and Conditional Value-at-Risk ($CVaR_{95}$ / Expected Shortfall) to assess extreme tail risk, complemented by the Ulcer Index for path-dependent drawdown stress.
- **Reverse Stress Threshold:** Solves for critical market shock ΔM^* such that $Loss(\Delta M^*) = L_{target}$, identifying the minimum systemic shock required to trigger portfolio breach.

6. Verification, Testing & Quality Assurance

To guarantee high reliability during judging and demonstration, the codebase underwent comprehensive automated testing and static verification:

Verification Suite	Scope & Coverage	Status / Result
Automated Python Test Suite	68 unit & integration tests covering quant math, contagion, router, storage, and auth	**PASSED (68/68 Green)**
Quantitative Engine Tests	29 dedicated math verification tests for shock calculus, VaR, and HHI	**PASSED (100% Green)**
API Contracts & Schemas	Pydantic v2 input validation, boundary tests, and error handling	**PASSED (100% Green)**
Frontend Build & Typecheck	TypeScript compilation and Vite production bundle optimization	**PASSED (Zero Errors)**
Frontend Code Quality	ESLint static code analysis across all UI components and pages	**PASSED (Zero Errors)**

7. Engineering Scope, Assumptions & Responsible AI

In accordance with the highest standards of financial technology ethics and the F-06 problem brief, the following operational scope is maintained:

- **Decision Support Scope:** Vertex Portfolio Risk Lab is an analytical decision-support and stress exploration demonstrator. It does not execute live trades or issue personalized financial, tax, or legal advice.
- **Explainable Quantitative Modeling:** Risk projections are generated from transparent quantitative formulas and sector sensitivity baselines rather than opaque predictive machine learning models, ensuring 100% auditable results.
- **Hypothetical Projection Guardrails:** The user interface explicitly distinguishes historical market baselines from forward-looking hypothetical projection cones to prevent any misconception of guaranteed returns.

8. Team Roles & Contributions

The project was designed, developed, and verified collaboratively by a multi-disciplinary team:

Member	Assigned Role	Key Contributions
Devireddy Nehal	Project Lead & Quant Engine	System architecture, quantitative calculation engine (quant.py), contagion network propagation (contagion.py), reverse stress testing, and test suite.
Mohammed Zama	Backend & API Engineering	FastAPI REST service orchestration, Pydantic data contracts, in-memory analysis caching, and database integration.
MD Ahmad	Frontend Lead	React/TypeScript application shell, page layouts, client-side API integration, and responsive state synchronization.
Shyam Sundar	Visualization & UI Widgets	Interactive correlation heatmap, historical vs. hypothetical chart widgets, scenario control sliders, and presentation deliverables.

9. Conclusion & Demonstration Highlights

Vertex Portfolio Risk Lab delivers an end-to-end, production-grade prototype for Problem F-06. By uniting interactive scenario exploration, transparent contagion modeling, analytical reverse stress solving, and intelligent guardrails, the platform empowers investors to look beyond isolated returns and truly understand their portfolio's hidden vulnerabilities. Supported by 68 passing automated tests, a fully typed codebase, and an intuitive modern UI, our submission stands ready for evaluation and live jury demonstration.